Lab 1: End-to-End Loan Approval Automation

Name: ABHINAV L | Register No.: RA2411003010553

Course: WORKFLOW MANAGEMENT

Camunda 8.8 | Java 17 | Spring Boot 3.5.6 | Synthetic data

Aim

Validate a synthetic application, retrieve a simulated credit score, evaluate documented eligibility rules, and route approval, rejection or human review.

Workflow steps

1. The starter supplies applicantRef, applicantName, age, loanAmount, monthlyIncome and creditProfile. All amounts are synthetic INR.

2. The loan-validate Java worker checks required fields, numeric types and positive values. Invalid input ends as Invalid application before credit/DMN evaluation.

3. The REST connector calls /mock/credit and stores creditScore and creditStatus. An HTTP error produces an incident instead of a business rejection.

4. The LoanDecision DMN table uses FIRST hit policy. Rejection rules take precedence, followed by strict approval and a manual-review fallback.

5. Approved and Rejected decisions reach terminal outcome workers. Manual/unexpected results create a Tasklist task assigned to reviewer (default demo).

6. The review form requires notes and an Approve/Reject decision. The gateway uses reviewDecision = "Approve"; all other values take the rejection route.

7. The record-outcome worker persists outcome and the process reaches the matching end event.

Rules, samples and assumptions

Hit policy: FIRST. The first matching rule returns one output.

Order	Condition	Output
1	age < 21	Rejected
2	monthlyIncome < 20000	Rejected
3	creditScore < 600	Rejected
4	loanAmount / monthlyIncome > 24	Rejected
5	age >= 21 AND income >= 30000 AND score >= 750 AND amount/income <= 12	Approved
6	Otherwise	Manual Review

Age; monthly income; credit score; loan amount	Expected result
30; 50000; 780; 300000	Approved
30; 50000; 500; 300000	Rejected
30; 50000; 650; 300000	Manual Review
21; 30000; 750; 360000	Approved

Validation

The automated test results record 39 JUnit tests, including 19 standalone DMN evaluations, plus 20 mock HTTP checks. Standalone rule tests used Camunda 7.24; live workflow screenshots separately demonstrate Camunda 8 process execution. Logs are retained under evidence/.

Assumptions

These are deliberately simple educational affordability rules, not lending policy. There is no loan disbursement or actual credit bureau lookup. An invalid request is distinguished from an eligible-format request rejected by DMN. Zero income is blocked before division.

Configuration and variables

README.md supplies startup/deployment commands. API_EXAMPLES.md documents REST mappings and request/response bodies. VARIABLE_DICTIONARY.md defines variable names, types and purposes. Editable models and all source code are included.

Test results

The workflows were tested on a Windows PC using Docker on 30-31 October 2026. The tables record the expected results, observed outcomes and process instance keys for the five required tests in each lab. Screenshots and copied process variables support these results. Extra example cases are listed separately and were not run.

Case / input JSON	Expected	Observed / instance	Status
L1-approved.json	Approved	Approved 2251799813686796	Passed
L2-rejected.json	Rejected	Rejected; copied variables plus completed-list evidence 2251799813691464	Passed
L3-manual.json	Manual loan review; Approved after reviewer approval	Manual Review -> reviewer Approve -> Approved 2251799813692297	Passed
L4-credit-error.json	Credit connector incident; Approved after fault cleared and retry	Credit incident -> correction/retry -> Approved 2251799813700233	Passed
L5-worker-error.json	Validation worker incident; Approved after fault cleared and retry	Validation worker incident -> correction/retry -> Approved 2251799813701734	Passed

Additional examples

L6-invalid, L7-manual-reject: supplied but not live-tested.

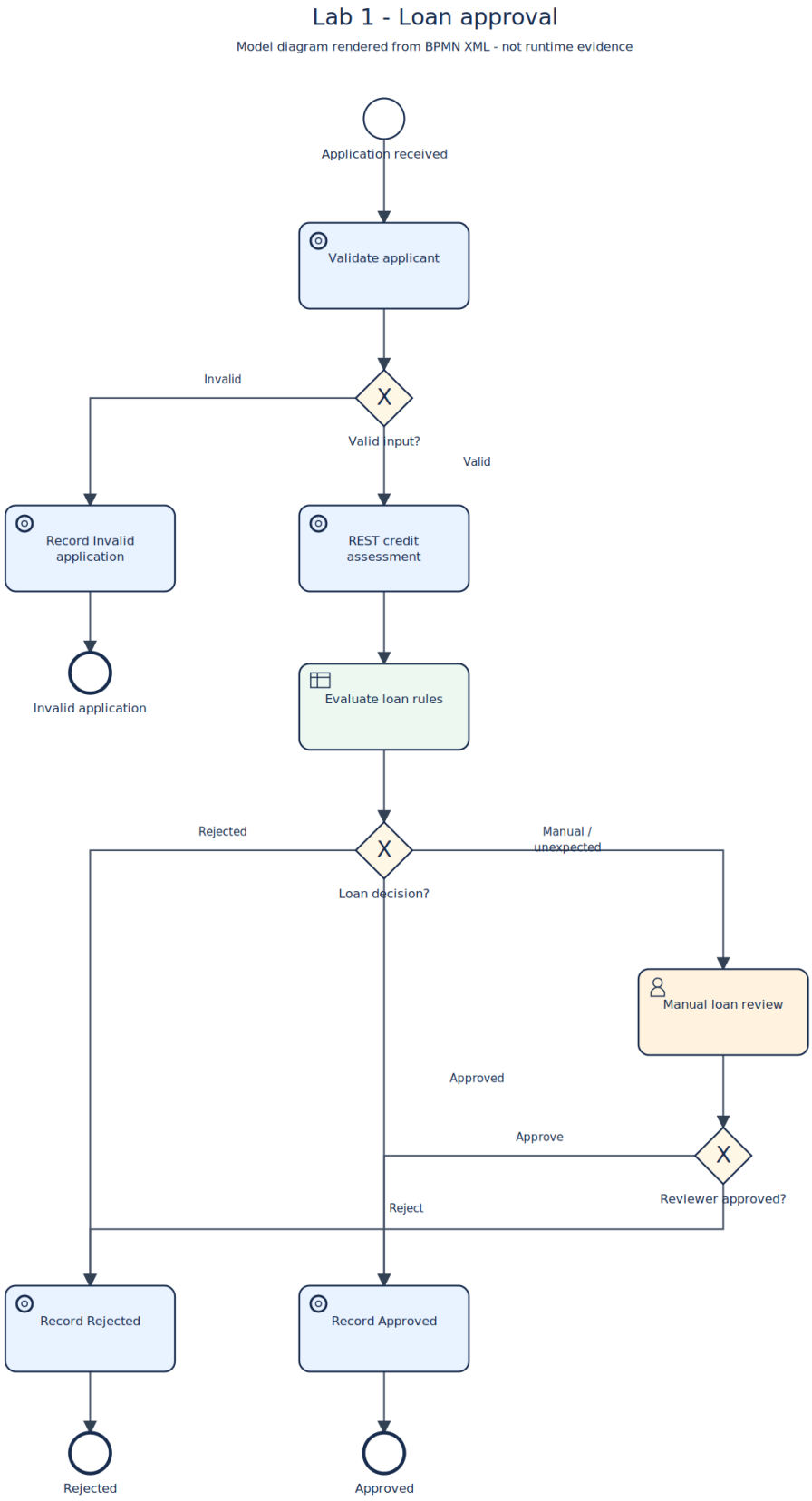
Notes on the evidence

The manual-review form capture is from duplicate instance 2251799813691797. The completed manual-review run is 2251799813692297. Rejection is supported by the completed instance list and copied variables.

Recovery procedure

Enable the case-specific mock fault, start the example and inspect the incident in Operate. Clear that fault and retry the failed job in the same instance. The failed external service returns HTTP 503; worker/connector failure exhausts retries and creates an incident. Loan worker faults are injected in validation. H5 reaches the scheduling API first through the availability worker. No real services or payments are involved.

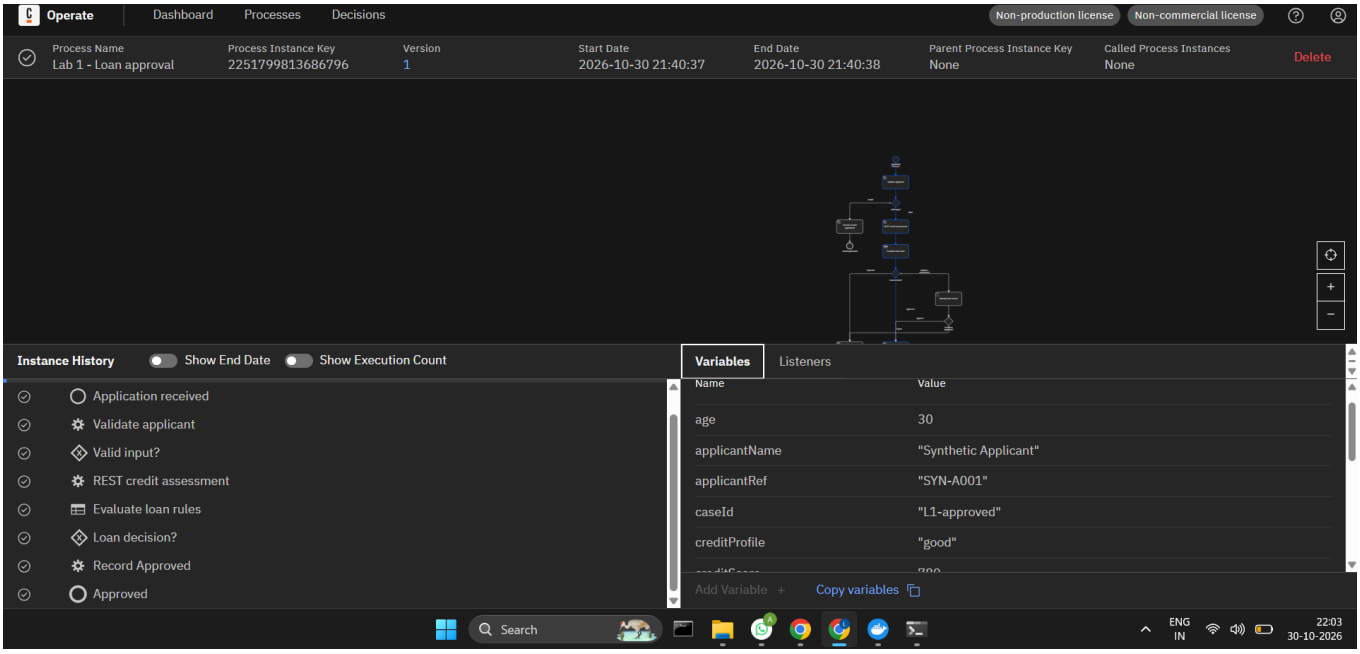
BPMN process design



Rendered from the supplied BPMN XML. Runtime captures follow; full-resolution original screenshots are in evidence/runtime/.

L1: approved

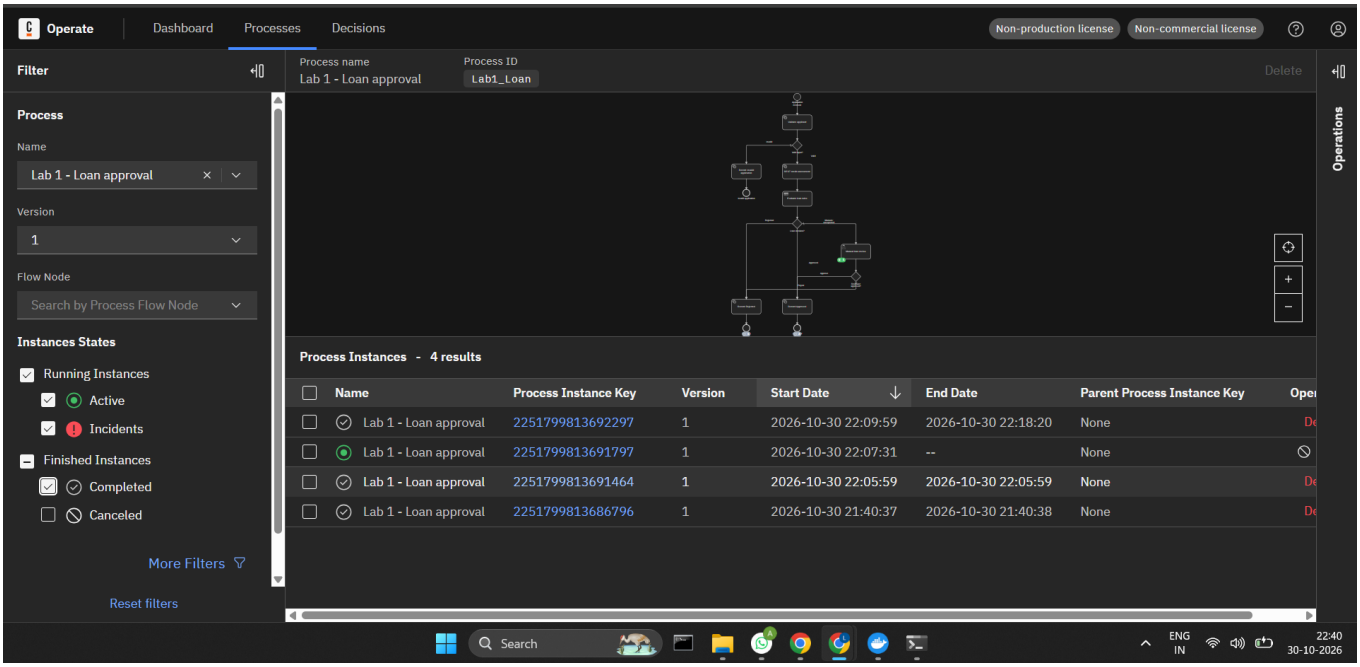
Approval route in Operate.



Full-resolution evidence: evidence/runtime/L1-approved.png

L2 list

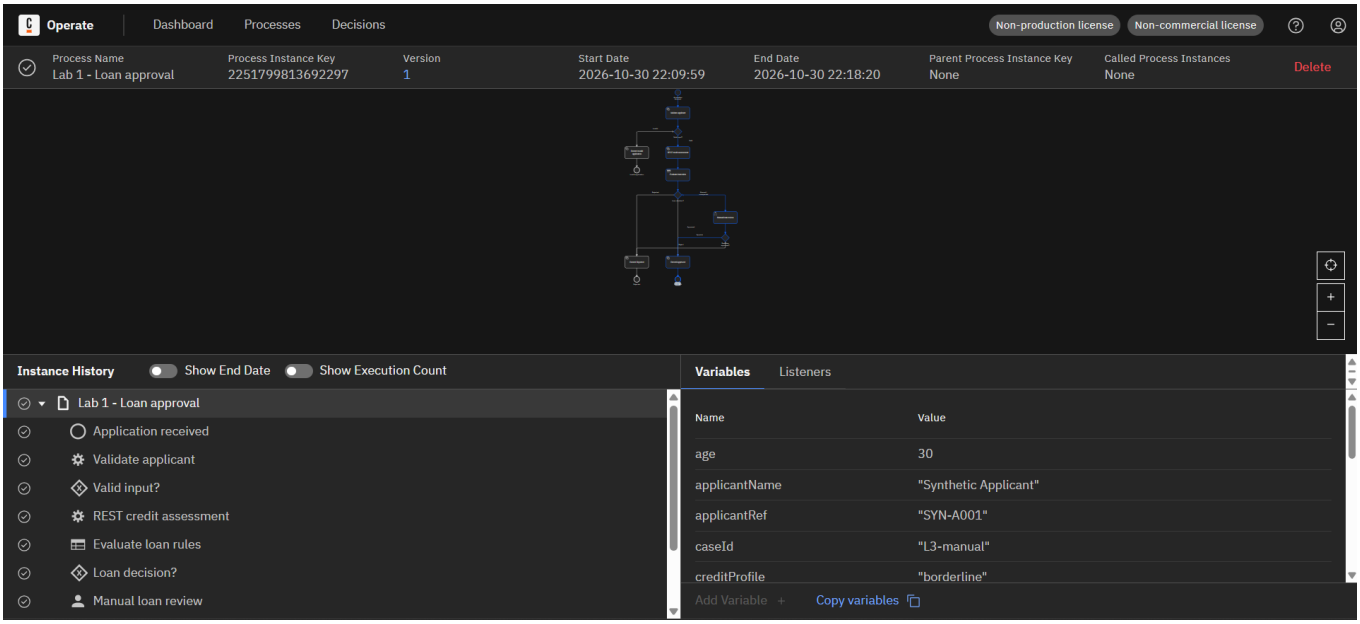
Completed loan instances; L2 key 2251799813691464. Rejected result also supplied as copied variables.



Full-resolution evidence: evidence/runtime/L2-list.png

L3 completed

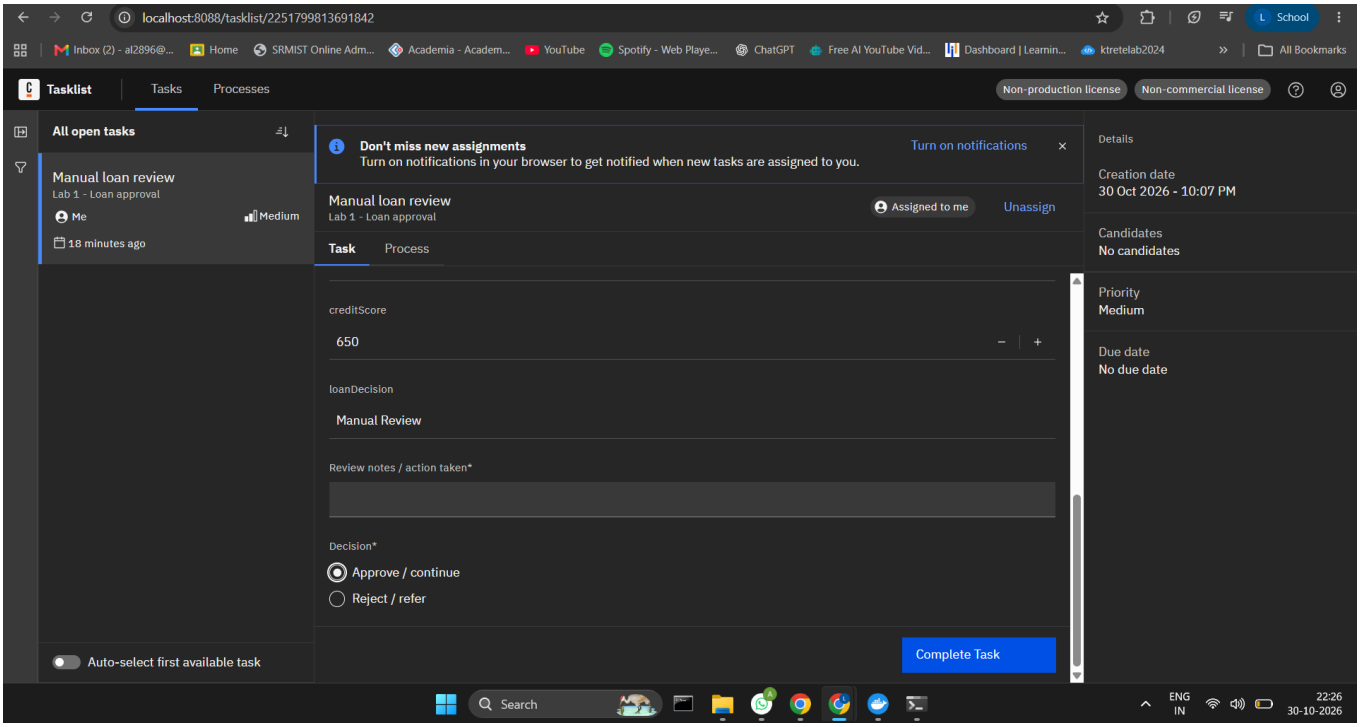
Completed manual-review instance 2251799813692297.



Full-resolution evidence: evidence/runtime/L3-completed.png

L3 task form

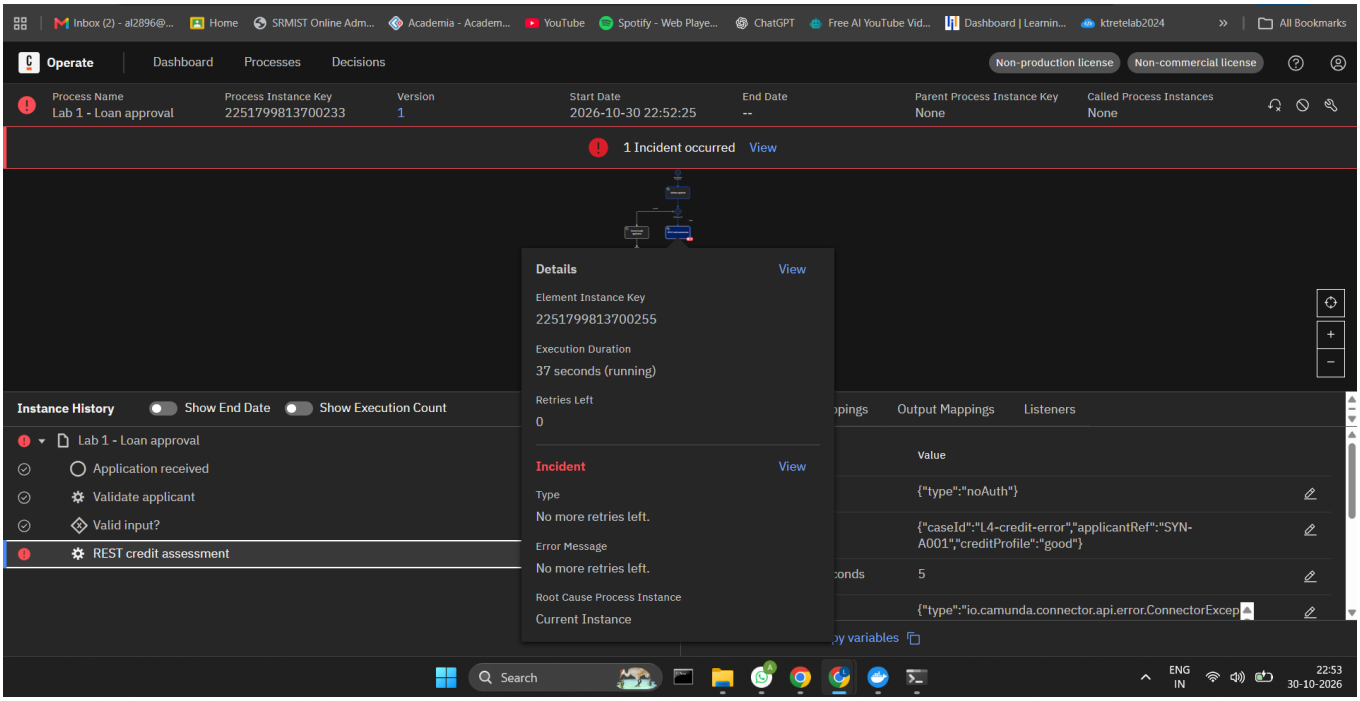
Manual-review form on duplicate instance 2251799813691797; form availability only, not submission proof.



Full-resolution evidence: evidence/runtime/L3-task-form.png

L4 incident

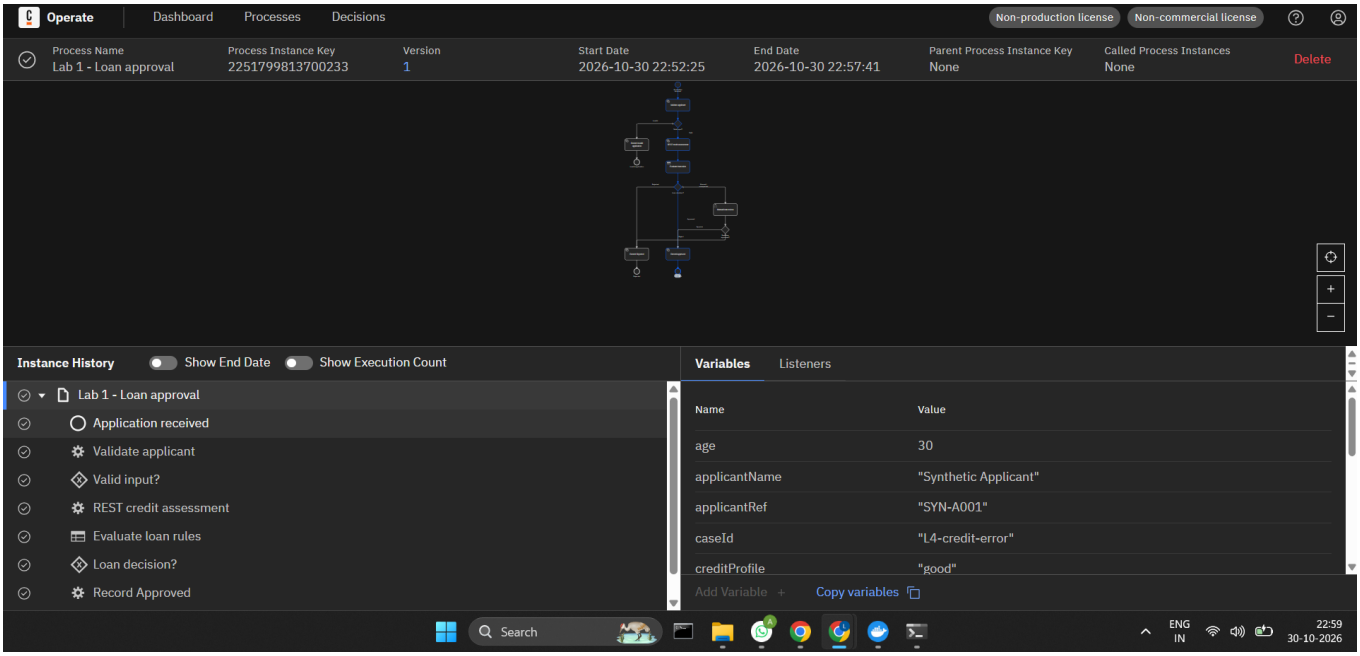
Credit connector incident; retries exhausted.



Full-resolution evidence: evidence/runtime/L4-incident.png

L4 recovered

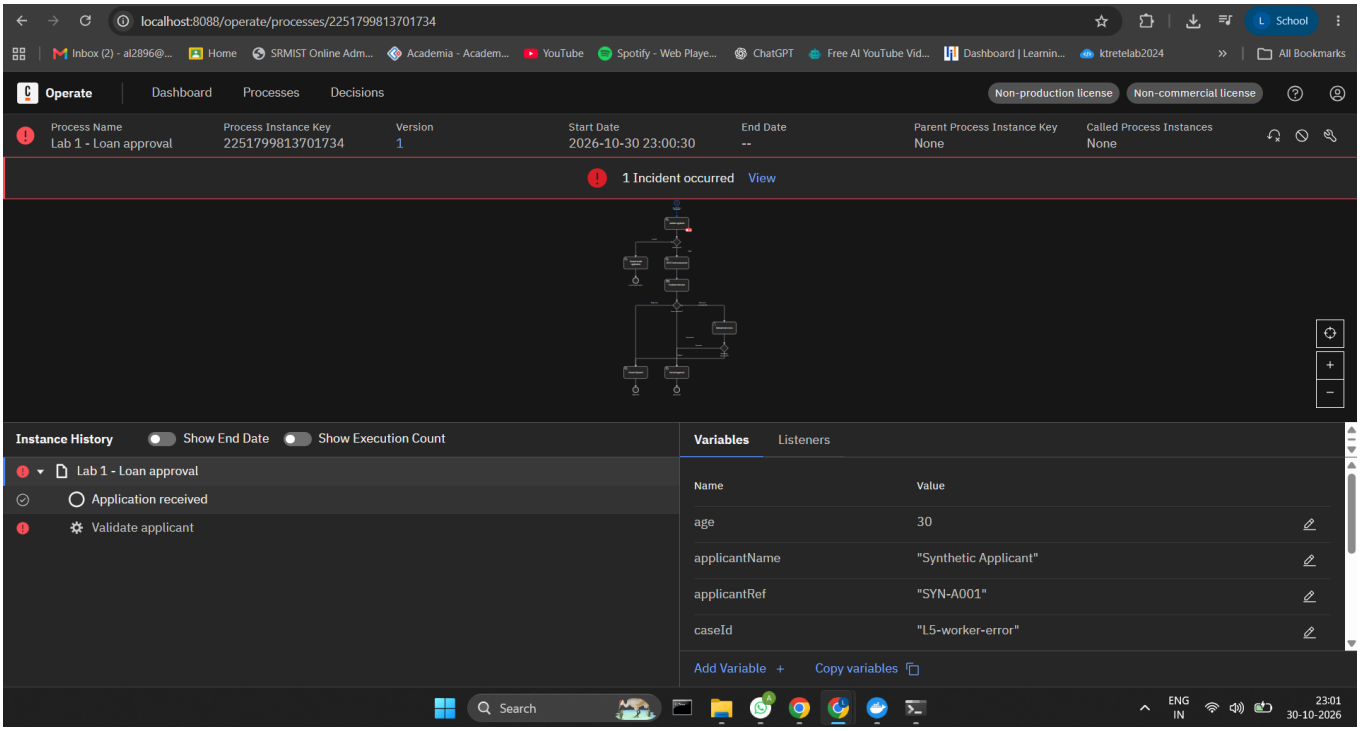
Same credit-error instance recovered and completed.



Full-resolution evidence: evidence/runtime/L4-recovered.png

L5 incident

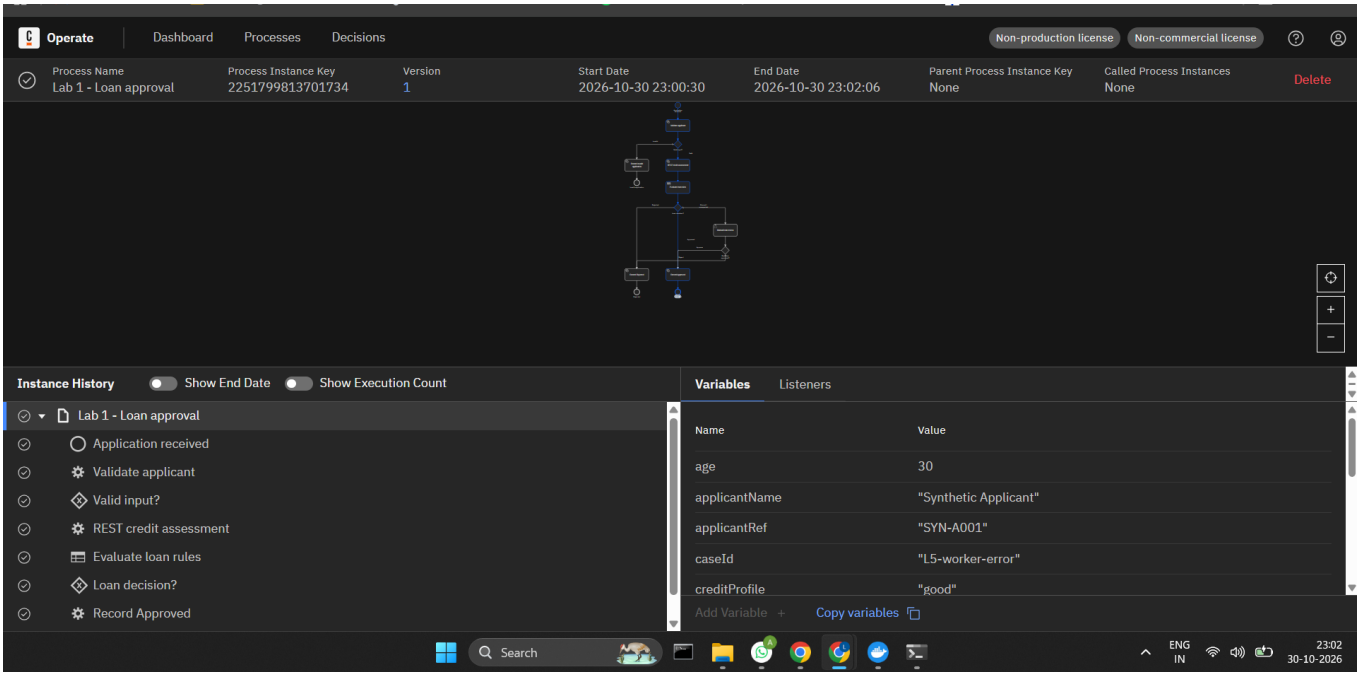
Validation worker incident.



Full-resolution evidence: evidence/runtime/L5-incident.png

L5 recovered

Same worker-error instance completed via Record Approved.



Full-resolution evidence: evidence/runtime/L5-recovered.png